

B.Tech. Degree VII Semester Regular/Supplementary Examination November 2023

ME 19-205-701 REFRIGERATION AND AIR CONDITIONING

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the basics of different refrigeration cycles.
 CO2: Study the vapour compression and vapour absorption refrigeration cycles- both theoretical and practical cycles, properties of refrigerants, and selection criteria for refrigerants.
 CO3: Understand the different components of refrigeration systems.
 CO4: Get knowledge of psychrometry, air-conditioning systems and will be able to do the cooling load calculation

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a)	How will you interpret 1 Tonne of refrigeration?	3	L1	1	1
(b)	Explain the principle of thermo-electric refrigeration	3	L1	1	1
(c)	Why do we use analyser and rectifier in practical vapour absorption system?	3	L1	2	1
(d)	What makes a refrigerant environment friendly? Give two examples of environment friendly refrigerants.	3	L1	2	1
(e)	Explain the working of a thermostatic expansion valve with the help of a neat sketch.	3	L1	3	1
(f)	What are the different factors affecting the condenser capacity?	3	L1	3	1
(g)	Define the following: (i) Specific humidity (ii) Absolute humidity (iii) Relative humidity.	3	L1	4	1
(h)	Explain room sensible heat factor and grand sensible heat factor. Show RSHF and GSHF lines in psychrometric chart.	3	L1	4	1

PART B

(4 × 12 = 48)

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|---------|---|---|----|---|---|
| II. (a) | Explain the boot - strap air evaporative cooling system. | 6 | L2 | 1 | 1 |
| (b) | A refrigerator working on Bell Coleman cycle operates between pressure limits of 1.05 bar and 8.5 bar. Air is drawn from the cold chamber at 10°C, compressed and then it is cooled to 30°C before entering the expansion cylinder. The expansion and compression follow the law $p v^{1.3} = \text{constant}$. Determine the theoretical coefficient of performance | 6 | L3 | 1 | 1 |

OR

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|----------|--|---|----|---|---|
| III. (a) | Explain the Carnot refrigeration cycle. Derive an expression for the coefficient of performance. | 6 | L2 | 1 | 1 |
| (b) | Explain the vortex tube refrigeration system with a sketch. Where it finds application? Why? | 6 | L3 | 1 | 1 |

(P.T.O.)

		Marks	BL	CO	PO										
IV.	(a) What are the desirable properties of an ideal refrigerant?	4	L1	2	1										
	(b) A refrigeration machine using R-12 as refrigerant operates between the pressures 2.5 bar and 9 bar. The compression is isentropic and there is no undercooling in the condenser. The vapour is in dry saturated condition at the beginning of the compression. Estimate the theoretical coefficient of performance. If the actual coefficient of performance is 65% of the theoretical value, calculate the net cooling produced per hour. The refrigerant flow is 5 kg per minute. Cp of superheated vapour at 9 bar as 0.64 kJ/kg K.	8	L3	2	2										
OR															
V.	(a) Make a comparative list between the different aspects of vapour absorption and vapour compression system.	4	L1	2	1										
	(b) Draw a neat diagram of practical aqua ammonia absorption system and explain its working.	8	L3	2	2										
VI.	(a) What are the advantages and disadvantages of centrifugal compressors over reciprocating compressors?	4	L2	3	1										
	(b) Explain the working of a single stationary blade type rotary compressor.	8	L2	3	1										
OR															
VII.	(a) Explain the operation of capillary tube in a refrigeration system.	6	L2	3	1										
	(b) With a neat sketch explain the working of flooded evaporators.	6	L2	3	1										
VIII.	(a) Air at 15°C DBT and 25% RH is heated and humidified to 30°C DBT and 50% RH. Calculate	4	L2	4	1										
	(i) The heat and moisture added to air														
	(ii) The sensible heat factor for the process.														
	(b) The following data is given for the space to be air conditioned:	8	L3	4	2										
<table border="1"> <tr> <td>Inside design conditions maintained:</td> <td>25°C DBT, 50% RH</td> </tr> <tr> <td>Outside air conditions:</td> <td>43°C DBT, 27°C WBT</td> </tr> <tr> <td>Room sensible heat gain:</td> <td>40 kW</td> </tr> <tr> <td>Room latent heat gain:</td> <td>10 kW</td> </tr> <tr> <td>By-pass factor of cooling coil used:</td> <td>0.2</td> </tr> </table>		Inside design conditions maintained:	25°C DBT, 50% RH	Outside air conditions:	43°C DBT, 27°C WBT	Room sensible heat gain:	40 kW	Room latent heat gain:	10 kW	By-pass factor of cooling coil used:	0.2				
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By-pass factor of cooling coil used:	0.2														
The return air from the room is mixed with outside air before entry to cooling coil in the ratio of 3:1 by weight. Determine:															
	(i) ADP of cooling coil.														
	(ii) Entry and exit conditions of air for cooling coil.														
	(iii) Dehumidified air quantity.														
	(iv) Fresh air mass flow rate and volume flow rate.														
	(v) Refrigeration load on the cooling coil (cooling plant).														
OR															
IX.	(a) Define effective temperature. Name the factors influencing effective temperature.	4	L2	4	1										
	(b) Air at dry bulb temperature of 30°C and 60% RH enters a cooling coil at the rate of 250 m ³ /min.	8	L3	4	2										
	(i) Determine the refrigeration in tonne needed to bring the temperature of air to 23°C through sensible cooling.														
	(ii) If the apparatus dew point temperature is 12°C and the bypass factor is 0.1, determine the refrigeration in tonne needed and the mass of water condensed out at the cooling coil per minute.														
	(iii) Determine the sensible heat factor for the process through the coil.														



